Jincheng Yang

Curriculum Vitae

Appointments

2025– **Assistant Professor**, *Johns Hopkins University*, Baltimore, MD

2024–2025 **Member**, *Institute for Advanced Study*, Princeton, NJ

2022–2024 **L.E. Dickson Instructor**, *University of Chicago*, Chicago, IL

2017 **Visiting Scholar**, *Georgia Institute of Technology*, Atlanta, GA

Education

2017–2022 **Ph.D. in Mathematics**, *The University of Texas at Austin*, Austin, TX
Advisor: Luis Caffarelli and Alexis Vasseur

2013–2017 **B.Sc. in Mathematics and Applied Mathematics** (Honors Program), *Xi'an Jiaotong University*, China

2011–2013 **Special Class for the Gifted Young**, *Xi'an Jiaotong University*, China

Research Interests

APDE **Analysis and Partial Differential Equations**, *dynamical systems, fluid dynamics, kinetic theory*
Euler equation, Navier–Stokes equation, Boltzmann equation, Fokker–Planck equation.

MSOR **Management Science and Operation Research**, *distributionally robust optimization, optimal mass transportation, risk measure*
Inventory problem, risk management, portfolio optimization.

Publications and Preprints

[1] Siming He, Yupei Huang, and Jincheng Yang. Fast separation of Euler and Navier–Stokes flows in two dimensions. *arXiv e-prints*, 2026. arXiv:2610.20540.

[2] Jacob Bedrossian, Jiajie Chen, Maria Pia Gualdani, Sehyun Ji, Vlad Vicol, and Jincheng Yang. Finite time singularities in the Landau equation with very hard potentials. *arXiv e-prints*, 2026. arXiv:2602.05981.

[3] Jincheng Yang, Vincent R. Martinez, Anna L. Mazzucato, and Alexis F. Vasseur. Energy dissipation near the outflow boundary in the vanishing viscosity limit. *Indiana University Mathematics Journal*, 75(4):1143–1176, 2026. arXiv:2410.13127, doi:10.1512/iumj.2026.75.241016.

[4] Jincheng Yang, Luhao Zhang, Ningyuan Chen, Rui Gao, and Ming Hu. Decision-making with side information: a causal transport robust approach. *Operations Research*, 74(4):2249–2269, 2026. OptOnline:20639. doi:10.1287/opre.2024.0997.

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[5] Matei P. Coiculescu and Jincheng Yang. Conditional Liouville theorems for the Navier–Stokes equations. *Calculus of Variations and Partial Differential Equations*, 65(1):22, 2026. arXiv:2506.14533, doi:10.1007/s00526-025-03196-y.

[6] Fei Cao and Jincheng Yang. Quantitative convergence guarantees for the mean-field dispersion process. *Discrete and Continuous Dynamical Systems*, 47:487–518, 2026. arXiv:2406.05043, doi:10.3934/dcds.2025126.

[7] Jincheng Yang. Vorticity interior trace estimates and higher derivative estimates via blow-up method. *Journal of Differential Equations*, 442:113486, 2025. arXiv:2308.09350, doi:10.1016/j.jde.2025.113486.

[8] Luhao Zhang, Jincheng Yang, and Rui Gao. A short and general duality proof for wasserstein distributionally robust optimization. *Operations Research*, 73(4):2146–2155, 2025. arXiv:2205.00362, doi:10.1287/opre.2023.0135.

[9] Alexis F. Vasseur and Jincheng Yang. Layer separation of the 3D incompressible Navier–Stokes equation in a bounded domain. *Communications in Partial Differential Equations*, 49(4):381–409, 2024. arXiv:2303.05236, doi:10.1080/03605302.2024.2346146.

[10] Luhao Zhang, Jincheng Yang, and Rui Gao. Optimal robust policy for feature-based newsvendor. *Management Science*, 70(4):2315–2329, 2024. OptOnline:18579. doi:10.1287/mnsc.2023.4810.

[11] Alexis F. Vasseur and Jincheng Yang. Boundary vorticity estimates for Navier–Stokes and application to the inviscid limit. *SIAM Journal on Mathematical Analysis*, 55(4):3081–3107, 2023. arXiv:2110.02426, doi:10.1137/22m1503567.

[12] Jincheng Yang. Construction of maximal functions associated with skewed cylinders generated by incompressible flows and applications. *Annales de l'Institut Henri Poincaré C. Analyse Non Linéaire*, 39(4):793–818, 2022. arXiv:2008.05588, doi:10.4171/aihpc/20.

[13] Alexis F. Vasseur and Jincheng Yang. Second derivatives estimate of suitable solutions to the 3D Navier-Stokes equations. *Archive for Rational Mechanics and Analysis*, 241(2):683–727, 2021. arXiv:2009.14291, doi:10.1007/s00205-021-01661-4.

[14] Zhiwu Lin, Jincheng Yang, and Hao Zhu. Barotropic instability of shear flows. *Studies in Applied Mathematics*, 144(3):289–326, 2020. arXiv:1801.00950, doi:10.1111/sapm.12297.

[15] Jincheng Yang and Zhiwu Lin. Linear inviscid damping for Couette flow in stratified fluid. *Journal of Mathematical Fluid Mechanics*, 20(2):445–472, 2018. arXiv:1610.08924, doi:10.1007/s00021-017-0328-3.

Honors and Awards

June 2026 AMS-Simons Travel Grant, *American Mathematical Society*

May 2022 Frank Gerth III Outstanding Dissertation Award, *UT Austin*

Mar 2021 University Graduate Continuing Fellowship, *UT Austin*

Apr 2019 Senate of College Council's TA of the Year, *UT Austin*

Invited Talks

Oct 2026	Center for Environmental and Applied Fluid Mechanics, *Johns Hopkins University*
Sep 2026	PDE Seminar, *Penn State University, University Park*
Sep 2026	PDE-Applied Math Seminar, *University of Maryland, College Park*
Aug 2026	Recent Breakthroughs and New Perspectives on the Interplay between Fluid Mechanics and Kinetic Theory, *Banff International Research Station*
Aug 2026	Partial Differential Equations for Many Particles Systems, *ICERM*
Jul 2026	2026 SIAM Annual Meeting (AN26), *Cleveland*
May 2026	Mathematical Developments in Geophysical Fluid Dynamics: Instabilities and transitions in geophysical flows, *Institut Henri Poincaré*
Dec 2025	PDE and Applications Seminar, *Chinese Academy of Sciences*
Dec 2025	PDE and Applications Seminar, *Chinese Academy of Sciences*
Oct 2025	AMS Fall Southeastern Sectional Meeting, *Tulane University*
Sep 2025	Applied Mathematics and Statistics Seminar, *Johns Hopkins University*
Aug 2025	AMS Fall Western Sectional Meeting, *University of Denver*
Jul 2025	ICCOPT 2025, *University of Southern California*
May 2025	2025 AWM Research Symposium, *University of Wisconsin, Madison*
May 2025	SIAM Conference on Applications of Dynamical Systems (DS25), *Denver*
Apr 2025	PDE and Analysis Seminar, *University of California, Los Angeles*
Apr 2025	Analysis Seminar, *The University of Texas at Austin*
Mar 2025	Analysis of Fluids and Related Topics, *Princeton University*
Nov 2024	Analysis and Mathematical Physics, *Institute for Advanced Study*
Nov 2024	Harmonic Analysis & PDE Seminar, *CUNY Graduate Center*
Oct 2024	Informs Annual Meeting 2024, *Seattle*
Oct 2024	PDE Seminar, *Brown University*
Oct 2024	Short Talks by Postdoctoral Members, *Institute for Advanced Study*
Aug 2024	Probabaility Seminar, *University of Illinois Urbana-Champaign*
Jun 2024	EquaDiff 2024, *Karlstad University*
May 2024	Recent Advances in Nonlinear Partial Differential Equations, *University of Minnesota*
May 2024	SITE Research Center Talk Series, *New York University Abu Dhabi (zoom talk)*
Mar 2024	AMS Spring Southeastern Sectional Meeting, *Florida State University*
Feb 2024	Simons Turbulence Seminar, *Johns Hopkins University (zoom talk)*
Nov 2023	Calderón-Zygmund Analysis Seminar, *University of Chicago*
Nov 2023	PDE Seminar, *Purdue University*
Oct 2023	Applied Mathematics and Statistics Seminar, *Johns Hopkins University*
Oct 2023	8th Annual Meeting of the SIAM Central States Section, *University of Nebraska-Lincoln*
Aug 2023	10th International Congress on Industrial and Applied Mathematics, *Waseda University (zoom talk)*
Jul 2023	PDE and Applications Seminar, *Chinese Academy of Sciences*

May 2023 Midwest PDE Seminar, *University of Notre Dame*
Jan 2023 Calderón-Zygmund Analysis Seminar, *University of Chicago*
Dec 2022 PDE and Applications Seminar, *Chinese Academy of Sciences (zoom talk)*
Nov 2022 Analysis of Fluids and Related Topics, *Princeton University*
Oct 2022 Informs Annual Meeting 2022, *Indianapolis*
May 2021 AMS Spring Western Sectional Meeting, *San Francisco State University (zoom talk)*
Mar 2021 AMS Spring Eastern Sectional Meeting, *Brown University (zoom talk)*

Service

Seminar Organizing

2023–2024 **Calderón–Zygmund Analysis Seminar**, *University of Chicago*
2019 **Harmonic Analysis Reading Seminar**, *University of Texas at Austin*

Journal Refereeing

SIAM Journal on Mathematical Analysis
Journal of Differential Equations
Mathematical Programming
Nonlinearity
Communications in Partial Differential Equations
Nonlinear Analysis
Journal of Machine Learning Research
Mathematische Annalen
Computers and Operations Research
SIAM Journal on Applied Mathematics
Bulletin of the Brazilian Mathematical Society
Acta Mathematica Scientia
Analysis & PDE

Teaching Experience

Instructor

Fall 2026 EN.553.701 – Real Analysis: Preparation for the Ph.D. Introductory Examination
Spring 2026 EN.553.746 – Stochastic Controls, Games, and Learning II
Fall 2025 EN.553.798 – Partial Differential Equations for Applied Mathematicians
Fall 2025 EN.553.101 – Freshman Experience in Applied Mathematics & Statistics
Spring 2024 Math 18500 – Mathematical Methods in the Physical Sciences III
Winter 2024 Math 27300 – Basic Theory of Ordinary Differential Equations
Spring 2023 Math 18500 – Mathematical Methods in the Physical Sciences III
Winter 2023 Math 20400 – Analysis in $\mathbb{R}^n$ II
Fall 2022 Math 18500 – Mathematical Methods in the Physical Sciences III

Teaching Assistant

Spring 2021	M383D – Methods of Applied Mathematics II
2019-2020	M427J – Differential Equations with Linear Algebra
Fall 2018	M427L – Advanced Calculus for Applications II
Spring 2018	M408D – Sequences, Series, and Multivariable Calculus
Fall 2017	M408K – Differential Calculus

Mentoring Undergraduate Students

Research Experience for Undergraduates

Summer 2024	Minh Pham, *on the topic of Nonlinear Integral Equations*
Summer 2024	Michael Lee, *on the topic of Partial Regularity Theory of Navier-Stokes Equations*
Summer 2024	Zichen Lu, *on the topic of Inviscid Damping of Euler Equations*

Reading Course

Spring 2024	Dante Strollo and Jakob Wellington, *on the topic of Fourier Analysis*

Directed Reading Program

Summer 2024	Ariana Qin, *on the topic of Stochastic Calculus and Stochastic Control*
Spring 2024	Ariana Qin, *on the topic of Optimal Control Theory*
Fall 2021	Kyle Alkire, *on the topic of Schauder Theory in Elliptic Equations*
Spring 2020	Yongqi Pang, *on the topic of Statistics and Data Analysis*
Spring 2019	Trey Minor, *on the topic of Differential Equations and Dynamical Systems*
Spring 2018	Yan Cheng, *on the topic of Probability and Martingales*